

CMPE 110 Audio Amplifier Project Report

This report documents the transition from the LTspice LM386 bridge amplifier simulation into a KiCad schematic, printed circuit board layout, 3D board model, and final assembled board. The design goal was a stereo audio amplifier using LM386 amplifier blocks with one volume control per channel and enough output swing to drive 4-ohm speakers. For a 2 W target into a 4-ohm load, the speaker requires about 2.83 V RMS, or about 8.0 V peak-to-peak across the load. The bridge configuration increases the voltage across the speaker by driving its terminals with opposite-polarity amplifier outputs.

1. LTspice bridge-amplifier simulation

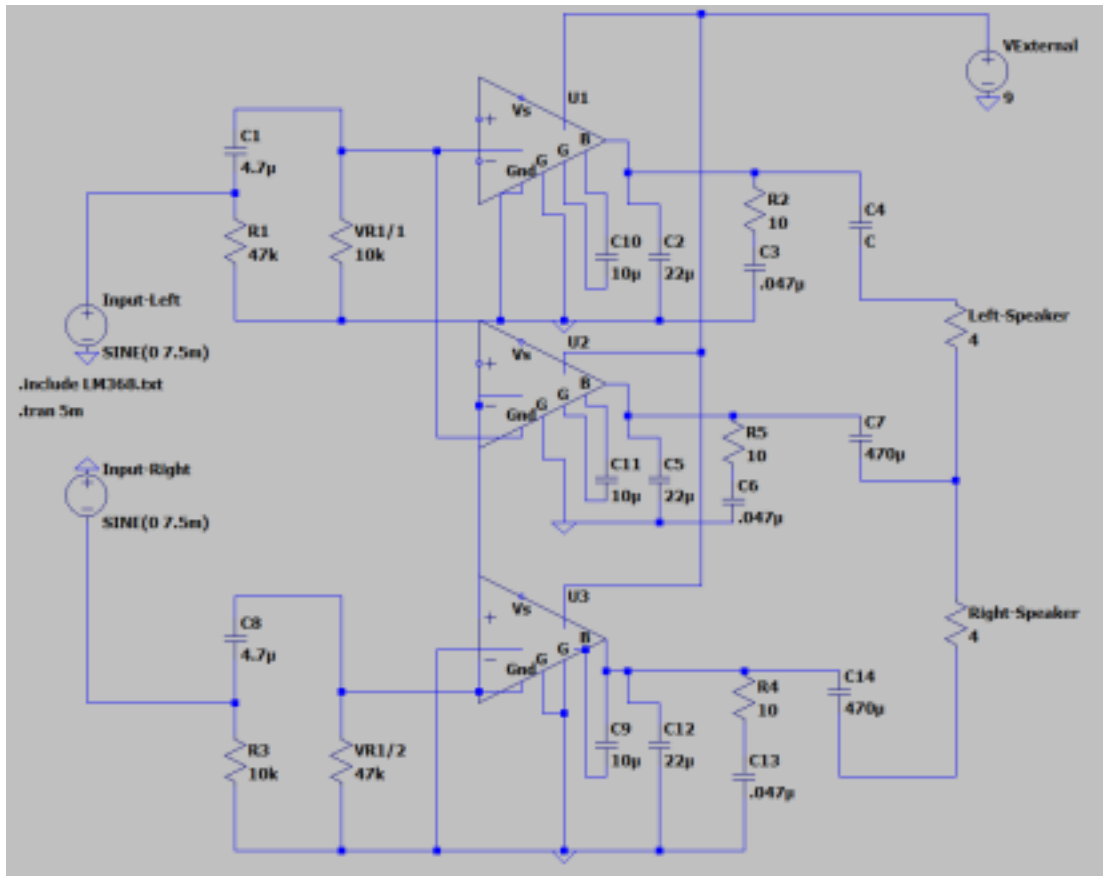


Figure 1. LTspice implementation of the LM386 stereo bridge amplifier circuit.

The LTspice circuit was used as the electrical reference before the physical design was created. It includes left and right audio input sources, coupling capacitors, gain-setting components, LM386 amplifier stages, output capacitors, and 4-ohm speaker loads. The left channel uses two LM386 stages in a bridged configuration so the speaker sees the difference between the two amplifier outputs, while the right channel is represented with its own LM386 output path. The potentiometer values model the volume controls, allowing the input signal level to be reduced or increased during testing.

The simulation verifies the expected behavior before any PCB work: the input audio signal is AC-coupled, the LM386 devices provide voltage gain, and the speaker load receives a larger output signal than the input.

2. KiCad schematic capture

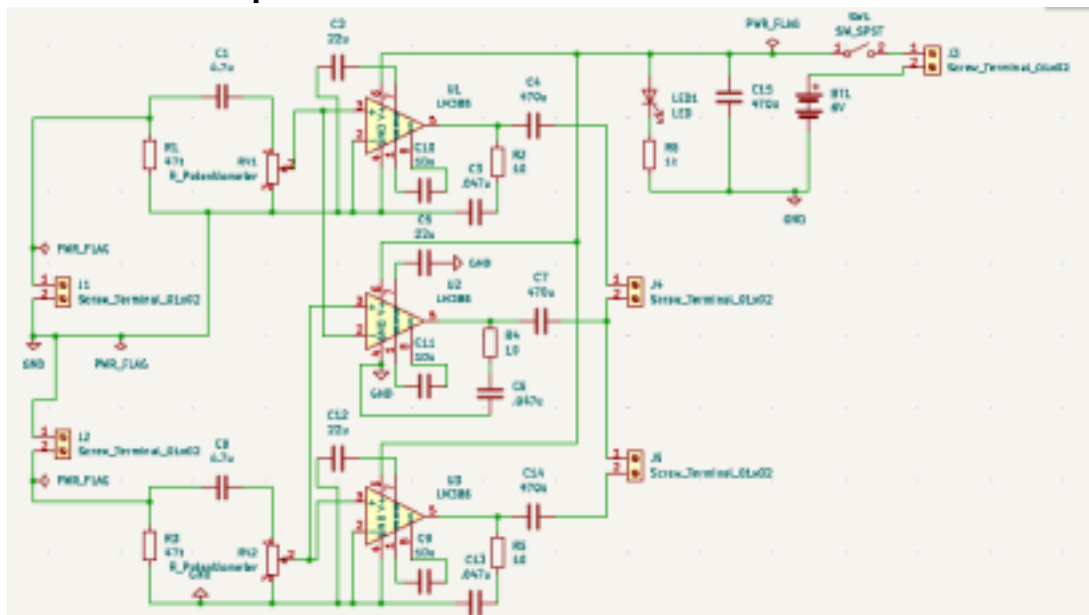


Figure 2. KiCad schematic translated from the LTspice circuit.

After the LTspice design was completed, the circuit was recreated in KiCad as a schematic suitable for PCB layout. The schematic keeps the same functional blocks: input terminals, two volume potentiometers, three LM386 ICs, coupling and bypass capacitors, speaker/output terminals, a battery input, a power switch, and an LED indicator. Each symbol is assigned to a physical footprint so the electrical design can be transferred directly into the PCB editor.

The schematic also makes the project easier to debug because every net has a defined purpose. The input nets connect to the potentiometers, the LM386 outputs connect through coupling capacitors and stability networks, and the supply rails connect to the battery, switch, and decoupling capacitors.

3. PCB layout with routed traces

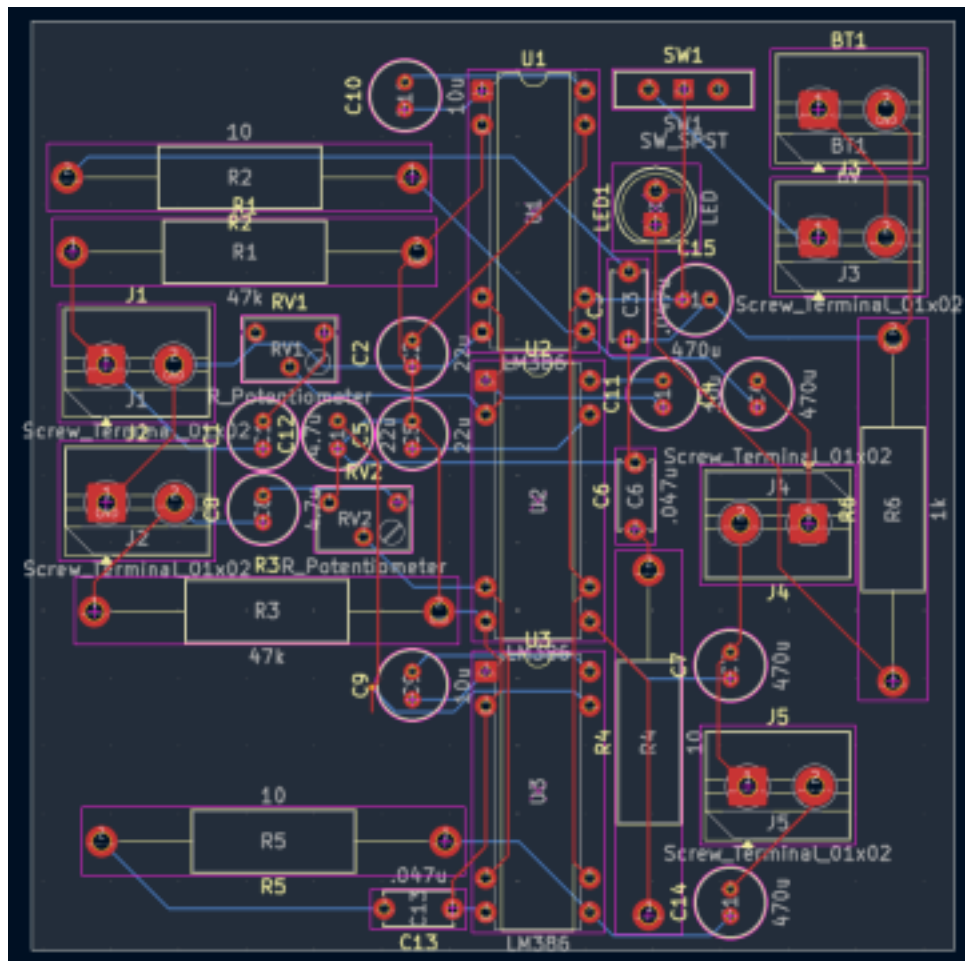


Figure 3. KiCad PCB layout showing component placement and routed copper traces.

The PCB layout converts the schematic into a manufacturable board. The larger through-hole components, including terminal blocks, potentiometers, DIP LM386 packages, electrolytic capacitors, resistors, and the switch, were arranged so that the input side, amplifier stages, power input, and speaker outputs are easy to identify. The routed traces connect the footprints according to the schematic netlist, and the board outline defines the final physical size.

The layout was organized to keep the audio path readable and serviceable. Input connectors and volume controls are grouped toward one side of the board, while the speaker/output connectors and power components are placed toward the opposite side

4. Design-rule check and layout verification

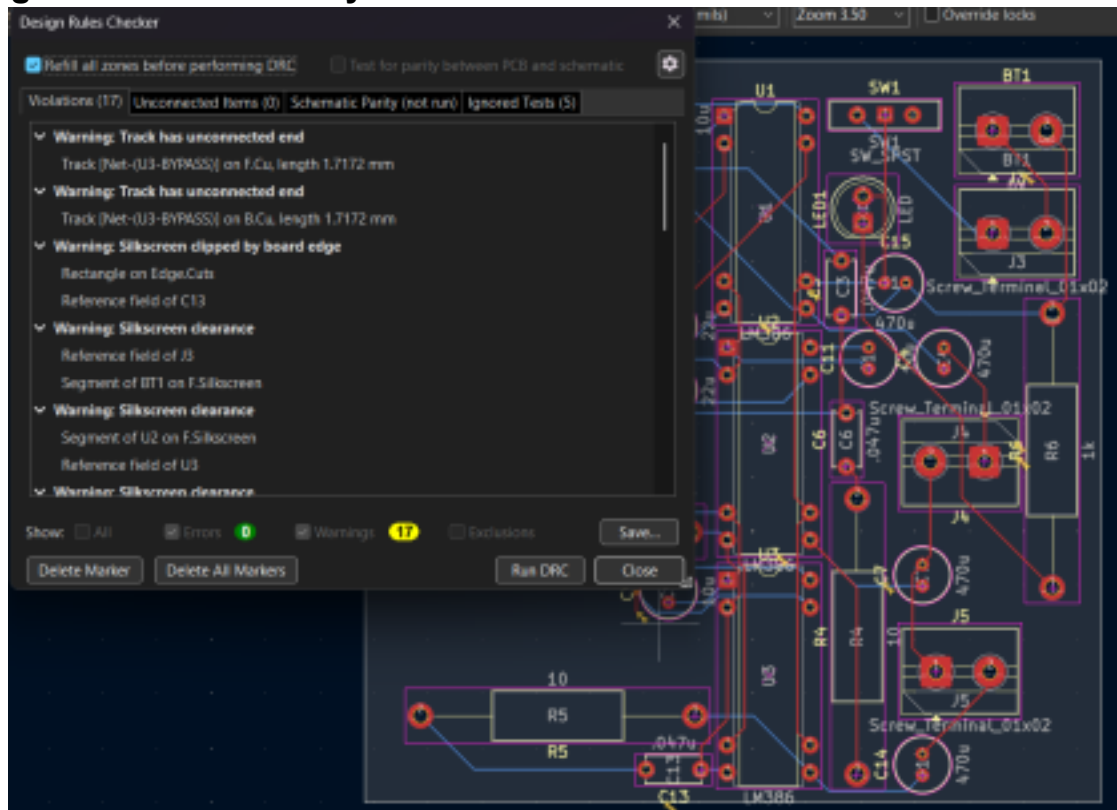


Figure 4. KiCad design-rule checker used to inspect the routed board.

Before manufacturing or assembly, the PCB was checked with KiCad design-rule checking. The DRC window identifies routing, clearance, and footprint issues that could cause fabrication problems or electrical faults. The visible warnings include unconnected tracks and silkscreen clearance warnings. These warnings were reviewed to determine whether they affected the actual circuit operation or were only cosmetic placement issues on the silkscreen layer.

This verification step is the bridge between the drawing stage and the build stage. Running DRC helps catch missing nets, shorts, traces too close to pads, and labels that overlap components.

5. 3D model of the PCB

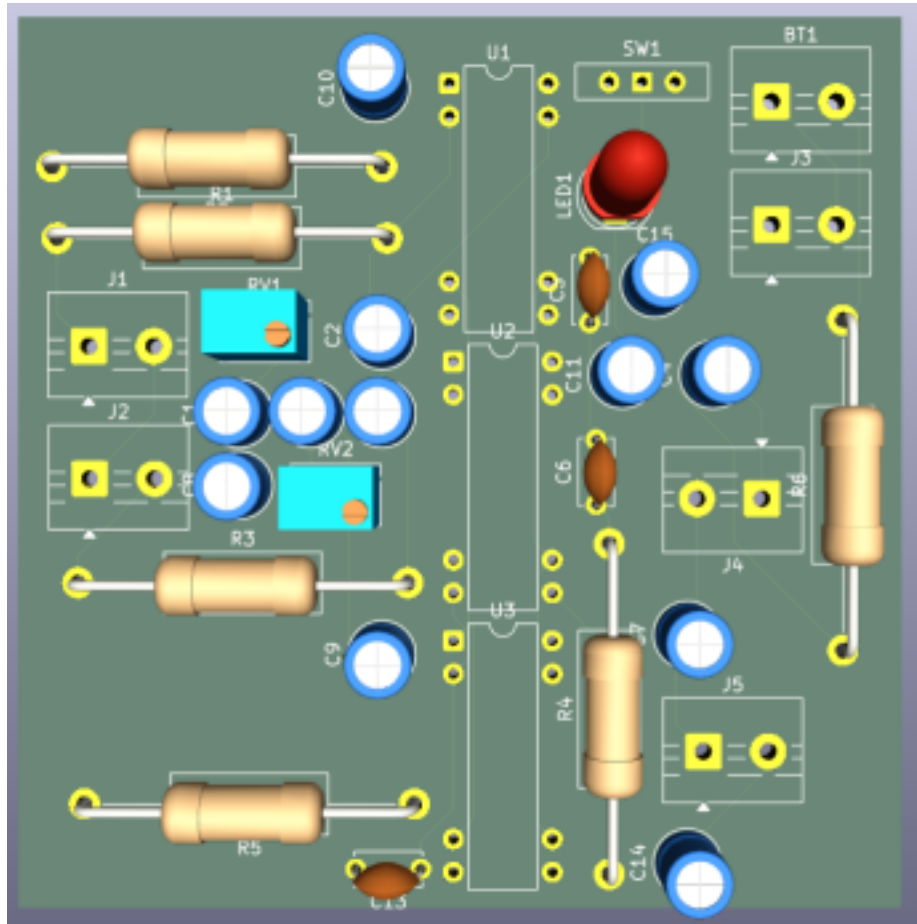


Figure 5. KiCad 3D viewer rendering of the audio amplifier board.

The 3D model provides a physical preview of the assembled board. It confirms that the selected footprints match the intended through-hole components and that parts such as capacitors, terminal blocks, potentiometers, resistors, the LED, and the switch have reasonable spacing. The 3D view also makes it easier to see whether tall components interfere with each other or with access to the screw terminals and adjustment knobs.

6. Fully built amplifier board



Figure 6. Completed audio amplifier project board after soldering and assembly.

The final board was assembled with the LM386 integrated circuits, resistors, capacitors, potentiometers, screw terminals, switch, LED, and mounting hardware installed. The physical build follows the KiCad layout and demonstrates that the simulated circuit was successfully transferred into a working hardware format. The two blue potentiometers provide the required user volume adjustments, and the screw terminals provide practical connections for input, power, and speaker wiring.

Overall, the project moved through the full design process: LTspice simulation, schematic capture, PCB routing, rule checking, 3D inspection, and physical assembly. The bridge amplifier concept is important because it increases the voltage across the speaker compared with a single-ended LM386 output, helping the design approach the required speaker power while still using LM386 amplifier blocks as required by the lab specification.

[Video showing the finished result.](#)